

AI Recommender Tool

Team members: Jake Carduck, Dylan Tran, Michelle Koh, Kelly Hwangpo | **Faculty adviser:** Caroline Budwell, Ph.D., John Leonard, Ph.D. | **Partner:** Capital One | **Mentor:** Nabin Debnath, Lakshmi Narayana Rasalay

Problem:

- All companies rely on a government, risk, and compliance (GRC) assessment for new job roles, projects, etc.
- The current process is:
 - Time consuming
 - Manual
 - Complex
 - Lacks scalability Goal

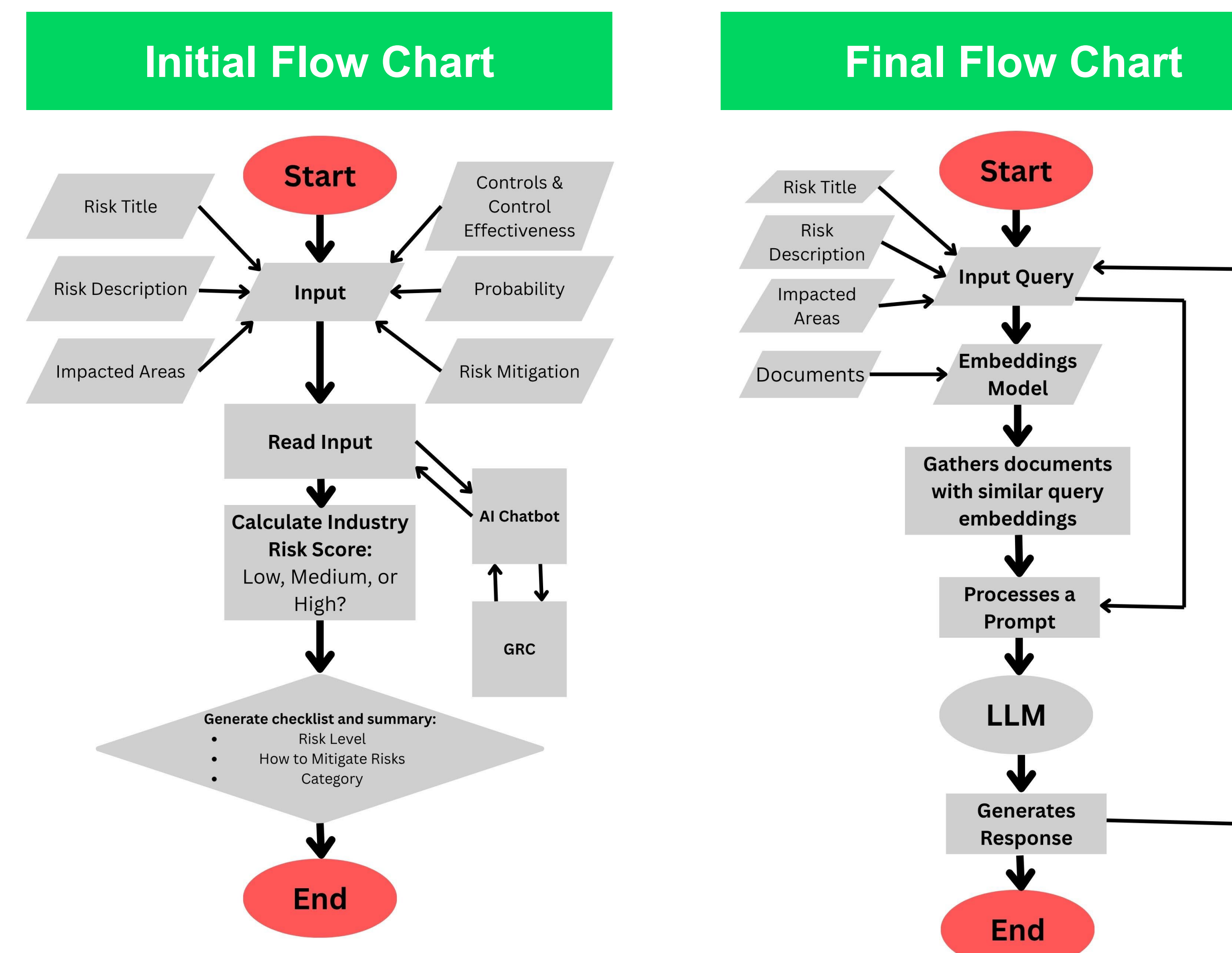
Goal:

- Simplify the risk management process, turning complex analysis into a straightforward, guided experience.

Impact:

- Automated risk assessment process for faster, more consistent results
- Improves efficiency and accuracy across risk evaluations
- Supports government, risk, and compliance standards
- Enhances decision-making using reliable, data-driven recommendations for risk mitigations
- Offers a user-friendly interface that simplifies the task
- Speeds up the overall assessment and mitigation overflow
- Uses machine learning to evolve and refine recommendations as more data becomes available

Flow Chart: Initial & Final Design



Challenges Faced:

- Time management
 - Scheduling and team coordination
- Finding the documents
 - Reliability
 - Accuracy
 - What documents can be used to indirectly train the AI for testing (since actual GRC data is proprietary)?
- Finding and running the right AI
 - Most offer few free tokens
 - Running the initial AI setup takes several minutes with our hardware and open-source AI for just a few documents

Technical Design Implementation:

- Document-driven RAG system (organization inputs documents)
- Cleans + normalizes text (PDF/Doc/etc.)
- Splits the text into chunks
- Generates embeddings for each chunk
- Stores embeddings + metadata in a vector database
- User inputs query → converts to an embedding
- Retrieves relevant chunks from vector database
- Builds context with the chunk selection
- LLM generates the answer using that context
- Responds with document references/citations

Languages:

- JavaScript
- HTML
- Java

What is GRC?

Emphasize governance, risk, and compliance ensures accountability in industrial industries

Frontend & UI

Risk Recommender Tool

Title
Enter a title...

Type
Risk

Description
This is a critical issue

Analyze

Risk Analysis:
Title:
Description: This is a critical issue
High

Risk Title Input

Type Input

Risk Description Input

Input Query to Embeddings Model

Risk Summary & Categorization